

VSEPR Bench

Why one molecule carries two shape names, and where the missing degrees go.

Name _____

Date _____

Period _____

Demonstration: andresdbp.github.io/resources/demos/vsepr-bench/

1 Before you open anything

Answer from what you already know and commit it to paper. The point is to have something concrete to compare against later, not to be right now.

PREDICTION

Water, H_2O , and sulfur dioxide, SO_2 , are both described as **bent**. Predict the bond angle in each one. Give a number for each, even if you are guessing.

H_2O : _____° SO_2 : _____°

Write one or two sentences saying how you arrived at those numbers.

2 Two names for one molecule

Open the demonstration. It starts on water. The panel *The central atom* has two sliders; the panel *The two names* – both of them reports the result twice.

1. Check that **Electron groups** reads 4 and **How many are lone pairs** reads 2. If not, press H_2O under *Real molecules*.
2. Fill in the first row of the table from the readouts.
3. Untick **Show lone pairs**. Fill in the second row from the same readouts, then tick it again.

Lone pairs showing?	Bonding groups	Electron-pair geometry	Molecular geometry
Yes			
No			

a. Hiding the lone pairs changed the picture. Did it change either name? Did any atom move?

b. The *Bond angle* panel reports an **Ideal** value and an **Actual** value. Record both, and copy the line printed underneath **Actual** word for word.

c. Copy the three values from **Repulsion ordering in this molecule**, in the order shown.

Which pairing gives the widest angle, and what does that tell you about how much room a lone pair takes compared with a bonding pair?

d. Tick **Show the ideal (uncompressed) bond positions**. Describe in one sentence where the faint dashed rays sit relative to the real bonds, and say which way water's bonds have been pushed.

3 What each lone pair costs

Now vary one thing at a time. Both sliders move in whole steps, so every setting below is a single click of the arrow keys once the slider has focus.

1. Set **Electron groups** to **4** and **How many are lone pairs** to **0**. Fill in that row.
2. Step the lone pairs up to **1**, then **2**, then **3**, filling in a row each time. Leave **Electron groups** at 4 throughout.
3. Now set **Electron groups** to **3** and repeat for **0** and **1** lone pairs.

Groups	Lone	Bonding groups	Electron-pair geometry	Molecular geometry	Ideal	Actual
4	0					
4	1					
4	2					
4	3					
3	0					
3	1					

a. Across the first four rows you held **Electron groups** at 4. Two of the columns you filled in stayed the same in all four rows. Name them, and say what those two quantities depend on.

b. Using rows 1 to 3, state the rule in your own words: what does each additional lone pair do to the bond angle, and roughly how many degrees does it cost on four groups? Then state how many degrees it costs on three groups.

c. The row with 4 groups and 3 lone pairs quotes no **Actual** angle at all. Read the line underneath and explain why not.

d. Two rows in your table carry the molecular geometry *bent*. Write both angles here, then go back to your prediction in Part 1. What did you assume that turned out not to hold?

4 Reading it the way a question would

Exam questions run this backwards: they hand you a shape or a formula and expect the count. Work each one out on paper first, then set the sliders and check.

a. A molecule is bent with a bond angle near 118° . How many electron groups surround the central atom, and how many are lone pairs? Set the sliders to your answer and copy the **Example** readout — both the formula and the line beneath it.

b. A molecule is described as AX_3E . Give its electron-pair geometry, its molecular geometry and its approximate bond angle without touching the demonstration, then check by setting the sliders. The AX notation is printed under the **Lone pairs** count.

c. Press CH_4 and read the caption underneath the molecule. It warns you about measuring the drawing with a protractor. What angle does the flat drawing read, what is the true angle, and why do the two differ?

d. Press SO_2 . The **Actual** readout and the **Example** readout do not quote the same number. Write both. Using the *Honest limits* panel, give one reason a measured angle can sit away from the value this model predicts.

5 Solve these without the demonstration

Close the page or look away. Draw the Lewis structure, count the electron groups, then answer. Show your working. Reopen the demonstration afterwards and check yourself — if a prediction missed, write down which step it went wrong at.

1. CCl_4 . Give the number of electron groups on the central carbon, the electron-pair geometry, the molecular geometry and the bond angle.

2. HCN, with carbon as the central atom. Give the number of electron groups, both geometries and the bond angle. Be explicit about how you counted the triple bond.

3. The nitrite ion, NO_2^- . Draw the Lewis structure, then give the electron-pair geometry, the molecular geometry and the approximate bond angle.

4. The H–C–H angle in CH_4 is 109.5° ; the H–N–H angle in NH_3 is about 107° . Both central atoms carry four electron groups. Explain the difference, and state the electron-pair geometry of each. No calculation — this one is answered in sentences.

6 On your own

Nothing here is graded, and the demonstration does more than this worksheet used.

- Press **Generic labels (A, X)** and work through every setting the two sliders allow – there are nine. Write the AX formula for each. Three of them refuse to quote a bond angle; find all three and say what they have in common.
- Untick **Show the electron-geometry frame** and tick it again at 3 groups and then at 4 groups. Watch what the frame is drawn between, and work out from the legend why some bonds are wedges and some are dashed.
- Read the *Honest limits* panel. H_2O is 104.5° but H_2S is 92.1° ; NH_3 is 107° but PH_3 is 93.5° . What information about the atoms does VSEPR never use, and which part of a prediction does it still get right?
- Tap each of the six shapes in the gallery and read the row headings above them. Which electron-pair geometry produces the most molecular geometries, and why that one?